# Stevão A. **Andrade**

Senior Software Engineering · Software Tester Expert

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## Research Interest

I work at the intersection of **Software Engineering** and **Software Testing**. My work draws on automate software testing activity, empirical research, mutation testing, and metamorphic testing to solve software engineering quality problems, such as fault detection, software testing oracle problem and software testing/engineering education.

## Education

### Universidade de São Paulo (USP)

*São Carlos, Brasil*

Ph.D. in Computer Science and Computational Mathematics Program — *Out. 2017 - Nov. 2022*

- Thesis: A software testing approach to virtual reality applications using metamorphic testing

### Universidade de São Paulo (USP)

*São Carlos, Brasil*

Master in Computer Science and Computational Mathematics Program — *Out. 2013 - Apr. 2016*

- Dissertation: Parallel execution of programs as a support for mutation testing

### Faculdade Integral Diferencial (FACID|DeVry)

*Teresina, Brasil*

Bachelor of Information Systems — *Jun. 2008 - jun. 2012*

- Undergraduate Thesis: An architecture to integrate GPS into web systems using Google Maps was supporting

## Professional Experience

### It's Elation, Inc. (Elation)

*Remote, USA*

Senior Software Engineer — *Jun. 2026 - Present*

- Technical lead of the SOC 2 Type 1 compliance effort (Thoropass), remediating AWS security findings, producing audit evidence, and addressing auditor requests;
- Hardening of the AWS network infrastructure, including NAT-routed private subnets, VPC Endpoints, a hardened bastion host, migration of Lambdas into the VPC, and least-privilege Security Groups;
- Migration of application secrets to AWS SSM Parameter Store and replacement of static IAM credentials with ECS task roles;
- Implementation of security monitoring and alerting with AWS GuardDuty, Security Hub (CIS and AWS FSBP standards), CloudTrail, AWS Config, CloudWatch, and EventBridge;
- Encryption at rest for AWS resources (e.g., EBS, SNS with customer-managed KMS keys) and enforcement of IAM password and MFA policies;
- Vulnerability management across back-end and front-end, including ECR image scanning, Dependabot remediation, Docker image hardening, and an Angular 18 to 21 upgrade;
- Authoring of security policies and compliance documentation (e.g., access control, risk assessment, vendor management, and incident response) aligned with the SOC 2 Trust Services Criteria;
- Back-end development with TypeScript/NestJS and Aurora MySQL, and validation of product features against functional specifications;

### Núcleo de Excelência em Tecnologias Sociais (NEES)

*Remote, Brasil*

Tech Lead – Development Team — *Jul. 2024 - Present*

- Leadership and technical guidance for a cross-functional development team;
- Design and implementation of scalable microservices architectures using AWS services (e.g., Lambda, SQS, RDS, and S3);
- Development and optimization of back-end services and APIs with PHP and Python;
- Implementation of asynchronous processing pipelines with message queues (e.g., Amazon SQS);
- Integration of monitoring and alerting tools (e.g., Grafana, Prometheus, and Testkube) to ensure system reliability;
- Collaboration with QA teams to define comprehensive test strategies and improve automation processes;
- Facilitation of Agile methodologies, ensuring efficient sprint planning and delivery using Jira and Confluence;

### Núcleo de Excelência em Tecnologias Sociais (NEES)

*Remote, Brasil*

Senior Test Engineer — *Jul. 2022 - Jul. 2024*

- Elaboration and execution of test scenarios;
- API testing using Karate, Postman and RestAssured;
- Exploratory testing;
- Web test automation using Selenium;
- Mobile test automation using Appium;
- Test and bug management and administration via ClickUp, Confluence and Jira;
- Development and extraction of reports for project and service quality metrics;
- Test monitoring using Testkube;
- Used tools: Testkube, Jira, Confluence, JMeter, K6, Postman, RestAssured, Grafana, Prometheus, GCP, Postman, and Appium.

**Universidade de São Paulo, USP** *São Carlos, Brasil*

Teacher Assistant *Fev. 2016 - Dez. 2019*

- Produce and teach content for undergraduate courses.
- Courses: Software Testing and Inspection, Object Oriented Programming, Introduction to Computer Science, Software Engineering.

**Universidade Virtual do Estado de São Paulo, UNIVESP** *São Paulo, Brasil*

Supervisor Professor *Jun. 2016 - Out. 2017*

- Monitor the virtual student activities according to the course schedule, guiding the teaching mediators in the publications in the forums and other activities.

**Conselho Regional de Administração do Piauí, CRA-PI** *Teresina, Brasil*

Information Technology Advisor *Jun. 2016 - Out. 2017*

- System development for process control; Build scripts to automate reporting; Monitoring and supervision tech interns; Built the institution's network infrastructure; Support and training to employees.
- Most used techs: Java, Python, MySQL, JEE, JSF, JPA, Ireport, JasperReporps.

**Tribunal Regional Eleitoral do Piauí, TRE-PI** *Teresina, Brasil*

Information Technology Intern *Jul. 2011 - Jan. 2012*

- Complementary system development for support reporting of project control tool Redmine; Maintenance of the institution Spending Control System.
- Most used techs: Java, JSF, Primefaces, JPA, Hibernate, JDBC, Ireport, JasperReporps.

**Conselho Regional de Administração do Piauí, CRA-PI** *Teresina, Brasil*

Information Technology Intern *Fev. 2010 - Fev. 2011*

- Operation of internal control systems; Support staff; Computer Maintenance; Backup routine control.
- Most used techs: Java, Python, MySQL

## Skills

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|---|---|
| **DevOps** | Docker, Kubernetes, Gitlab CI |
| **Back-end** | Springboot, Django, REST API |
| **Front-end** | HTML5, Javascript |
| **Programming** | Java, Python, C#, C, LaTeX |
| **Languages** | Portuguese, English (B2) |
| **Testing** | Testkube, Selenium, JUnit, Infer, Pit, Pytest, Evosuite, Randoop, PMD, SpotBugs, JMeter, K6, Postman |

## Extracurricular Activity

**Software Engineering Laboratory (LabES)** *Brasil*

Member *Jan. 2013 - Present*

- Gained expertise in software engineering field developing a wide range of research with several academic papers, focusing especially on software testing, software testing education and empirical software engineering.
- Participated on several academic conferences.

**Open Source Development** *Github*

Enthusiastic *2015 - Present*

- Part time contributor to some open source community projects on *Github*.

**Força Junior Informática** *Teresina, Brasil*

Member *Jan. 2009 - Jun. 2012*

- Undergraduate junior company. Gained expertise in software development developing several web applications such as voting system, student control system, process control system.

## Honors & Awards

### International

2020 **Finalist**, Mentor recognition of excellence in mentoring on Technovation Girls program. *Online, U.S.A*

2019 **First Place**, IEEE International Software Testing Contest, IEEE society. *Salvador, Brasil*

### Domestic

2019 **Recognition**, One of the 8 candidates for the Best Paper prize of SVR 2019. *Rio de Janeiro, Brasil*

2011 **Recognition**, Academic Merit for performance in the undergraduate Information Systems course *Teresina, Brasil*

# Program Committees

## Journals

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| 2023 | **Technical Reviewer**, Software Testing, Verification and Reliability Journal | *Online* |
| 2022 | **Technical Reviewer**, International Journal of Software Engineering and Knowledge Engineering | *Online* |

## Conferences

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| 2024 | **Technical Reviewer**, 9th Brazilian Symposium on Systematic and Automated Software Testing | *Curitiba, Brasil* |
| 2023 | **Technical Reviewer**, 8th Brazilian Symposium on Systematic and Automated Software Testing | *Campo Grande, Brasil* |
| 2023 | **Technical Reviewer**, 16th IEEE International Conference on Software Testing, Verification and Validation | *Dublin, Ireland* |
| 2023 | **Technical Reviewer**, 8th International Workshop on Metamorphic Testing | *Melbourne, Australia* |
| 2022 | **Technical Reviewer**, 7th International Workshop on Metamorphic Testing | *Pittsburgh, USA* |
| 2022 | **Technical Reviewer**, 7th Brazilian Symposium on Systematic and Automated Software Testing | *Uberlandia, Brasil* |
| 2021 | **Technical Reviewer**, 6th International Workshop on Metamorphic Testing | *Madrid, Spain* |
| 2021 | **Technical Reviewer**, 6th Brazilian Symposium on Systematic and Automated Software Testing | *Joinville, Brasil* |
| 2020 | **Technical Reviewer**, 5th Brazilian Symposium on Systematic and Automated Software Testing | *Natal, Brasil* |
| 2020 | **Technical Reviewer**, 34th Brazilian Symposium on Software Engineering | *Natal, Brasil* |
| 2020 | **Technical Reviewer**, 10th A-TEST workshop | *Singapore* |
| 2019 | **Technical Reviewer**, Simpósio Iniciação Científica e Tecnológica da Universidade de São Paulo | *São Carlos, Brasil* |
| 2017 | **Technical Reviewer**, 1st Simpósio de Engenharia de Software | *Dois Vizinhos, Brasil* |

# Academic Publications (Chronological Order)

## Refereed Journal Articles

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|---|---|---|
| 2023 | **ACM Surveys** 🕮, **ANDRADE, STEVÃO A.** ; NUNES, FATIMA LS ; DELAMARO, MÁRCIO . *Exploiting deep reinforcement learning and metamorphic testing to automatically test virtual reality applications*. Software Testing, Verification and Reliability 33 (8), e1863. | *doi* |
| 2022 | **ACM Surveys** 🕮, JÚNIOR, MISAEL C. ; AMALFITANO, DOMENICO ; GARCÉS, LINA ; FASOLINO, ANNA RITA ; **ANDRADE, STEVÃO A.** ; DELAMARO, MÁRCIO . *Dynamic Testing Techniques of Non-Functional Requirements in Mobile Apps: A Systematic Mapping Study*. ACM COMPUTING SURVEYS, v. 1, p. 3507903, 2022. | *doi* |
| 2022 | **REIC** 🕮, BRITO, R. O. ; **ANDRADE, STEVAO A.** ; DELMARO, M. E. . *Uma Avaliação de técnicas e Critérios de Teste de Software para a Linguagem de Programação Python*. REVISTA ELETRÔNICA DE INICIAÇÃO CIENTÍFICA, v. 20, p. 1-16, 2022. | *doi* |
| 2021 | **IJSEKE** 🕮, DELAMARO, MÁRCIO E. ; **ANDRADE, STEVÃO A.** ; DE SOUZA, SIMONE R. S. ; DE SOUZA, PAULO S. L. . *Parallel Execution of Programs as a Support for Mutation Testing: A Replication Study*. INTERNATIONAL JOURNAL OF SOFTWARE ENGINEERING AND KNOWLEDGE ENGINEERING, v. 31, p. 337-380, 2021. | *doi* |
| 2019 | **CI** 🕮, **ANDRADE, STEVÃO A.**; L. S. NUNES, FATIMA ; E. DELAMARO, MARCIO . *A fault-based testing approach for VR applications*. Comunicações em Informática, v. 3, p. 1-4, 2019. | *doi* |

## Book Chapters

| | | |
|---|---|---|
| 2018 | 🗐, OLIVEIRA, R. A. P. ; SOUZA, F. C. M. ; **ANDRADE, S. A.** ; MACIEL, A. C. ; FERRARI, F. C. ; DELAMARO, M. . *Ferramentas de Teste de Mutação*. In: Jose Maldonado, Marcio Delamaro, Auri Marcelo Rizzo Vincenzi. (Org.). Automatização de teste de software com ferramentas de software livre. 1ed.: Elsevier Brasil, 2018, v. , p. 1-256. | *doi* |
| 2018 | 🗐, Souza, Francisco Carlos ; Santos, Alinne ; **Andrade, Stevão** ; Durelli, Rafael ; Durelli, Vinicius ; Oliveira, Rafael . *Automating Search Strings for Secondary Studies*. Advances in Intelligent Systems and Computing. 1ed.: Springer International Publishing, 2018, v. , p. 839-848. | *doi* |
| 2016 | 🗐, COSTA JUNIOR, M. ; **ANDRADE, S. A.** ; DELAMARO, Márcio Eduardo . *Automatização de teste de software com ênfase em teste de unidade*. In: Universidade Federal do Piauí. (Org.). II Escola Regional de Informática do Piauí: livro texto dos minicursos. 1ed.Teresina: Sociedade Brasileira de Computação, 2016, v. 1, p. 121-143. | *doi* |

## Refereed Conference Papers (Research Track)

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| 2025 | **WBOTS**, VILELA, RICARDO F. ; **ANDRADE, STEVÃO A.** ; SANTOS, EDUARDO B. F. ; SILVA, WILLIAMSON ; VALLE, PEDRO H. D. . *An Intelligent Agent for Automated Test Generation from OpenAPI Specifications*. In: WBOTS 2025: 2nd Workshop on Bots in Software Engineering, 2025, Recife Brazil. Anais do Workshop sobre Bots na Engenharia de Software (WBOTS), 2025. p. 67-76. | *doi* |
| 2020 | **SAST**, SANTOS, SEBASTIÃO H. N. ; DA SILVEIRA, BEATRIZ NOGUEIRA CARVALHO ; **ANDRADE, STEVÃO A.** ; DELAMARO, MÁRCIO ; SOUZA, SIMONE R. S. . *An Experimental Study on Applying Metamorphic Testing in Machine Learning Applications*. In: SAST 20: 5th Brazilian Symposium on Systematic and Automated Software Testing, 2020, Natal Brazil. Proceedings of the 5th Brazilian Symposium on Systematic and Automated Software Testing, 2020. p. 98. | *doi* |
| 2020 | **SVR**, **ANDRADE, STEVAO A.**; QUEVEDO, ALVARO JOFFRE U. ; NUNES, FATIMA L. S. ; DELAMARO, MARCIO E. . Understanding VR Software Testing Needs from Stakeholders' Points of View. In: 2020 22nd Symposium on Virtual and Augmented Reality (SVR), 2020, Porto de Galinhas. 2020 22nd Symposium on Virtual and Augmented Reality (SVR), 2020. p. 57. | *doi* |
| 2019 | **MET**, **ALVES DE ANDRADE, STEVAO**; SANTOS, ITALO ; BRITO JUNIOR, CLAUDINEI ; JUNIOR, MISAEL ; R.S. DE SOUZA, SIMONE ; E. DELAMARO, MARCIO . *On Applying Metamorphic Testing: An Empirical Study on Academic Search Engines*. In: 2019 IEEE/ACM 4th International Workshop on Metamorphic Testing (MET), 2019, Montreal. 2019 IEEE/ACM 4th International Workshop on Metamorphic Testing (MET), 2019. p. 9. | *doi* |
| 2019 | **SAST**, **ANDRADE, STEVÃO A.**; BRITO, CLAUDINEI ; JÚNIOR, MISAEL ; MARCIEL, ANA CLAUDIA ; ABDALLA, GABRIEL ; DELAMARO, MÁRCIO E. . *Analyzing the effectiveness of One-Op Mutation against the minimal set of mutants*. In: the IV Brazilian Symposium, 2019, Salvador. Proceedings of the IV Brazilian Symposium on Systematic and Automated Software Testing - SAST 2019, 2019. p. 22. | *doi* |
| 2019 | **SBES**, DE ANDRADE, STEVÃO ALVES; DE OLIVEIRA NEVES, VÂNIA ; DELAMARO, MÁRCIO EDUARDO . *Software Testing Education*. In: the XXXIII Brazilian Symposium, 2019, Salvador. Proceedings of the XXXIII Brazilian Symposium on Software Engineering - SBES 2019, 2019. p. 47. | *doi* |
| 2019 | **FIE**, AVELLAR, GUSTAVO M. N. ; SILVA, ROGERIO F. DA ; SCATALON, LILIAN P. ; **ANDRADE, STEVAO A.** ; DELAMARO, MARCIO E. ; BARBOSA, ELLEN F. . *Integration of Software Testing to Programming Assignments: An Experimental Study*. In: 2019 IEEE Frontiers in Education Conference (FIE), 2019, Covington. 2019 IEEE Frontiers in Education Conference (FIE), 2019. p. 1. | *doi* |
| 2019 | **SVR**, **ANDRADE, STEVAO A.**; NUNES, FATIMA L. S. ; DELAMARO, MARCIO E. . *Towards the Systematic Testing of Virtual Reality Programs*. In: 2019 21st Symposium on Virtual and Augmented Reality (SVR), 2019, Rio de Janeiro. 2019 21st Symposium on Virtual and Augmented Reality (SVR), 2019. p. 196. | *doi* |